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a.

b.

Figure 10.30: Shows grain, texure and figure of wood (a–b).

Activity

Collect some pieces of plywood, analyse the layers and discuss yourself how it is made.

Summary

Secondary growth deals with the formation of additional vascular tissue by the activities of vascular and cork cambia and secondary thickening meristem (STM). It increases the girth of stem and roots of gymnosperms, most angiosperms, and some monocot plants. Vascular cambium possesses two kinds of initials they are, fusiform and ray initials. Fusiform initials give rise to the axial tissue system whereas ray initials give rise to radial tissue system of stems and roots.

Wood is a very important product of secondary growth. It represents secondary xylem. It is classified in various ways. Based respectively on the presence or absence of vessels, wood is classified into two types. i.e., porous and non-porous wood. Based on the wood formed during seasons, it is classified into spring wood and autumn wood. The spring and autumn wood, together is called **annual ring**. The wood is also classified into sap wood (pale in colour) and heart wood (dark in colour). The lumen of the xylem vessels of heart wood are blocked by many balloon like ingrowths from neighbouring parenchymatous cells called **tyloses**.

The periderm, a secondary protective tissue consists of phellem, phellogen and phelloderm. Secondary growth produces a corky bark around the tree trunk that protects the interior parts from heat, cold, infection etc. Secondary growth of root is different from stem in the method of formation of vascular cambium.

Anomalous secondary growth is now referred as cambial variants. They are abnormal types of secondary growth that occur in some dicots and monocots. Timber is derived from wood logs. In order to enrich the quality of timber, seasoning of wood is done through air and kiln drying. Wood is characterized by colour, grain, texture and figure.